P05

A General Method for Sequence Similarity with Gap Penalties

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Domain DOM_BIO

Abstract

A dynamic programming approach is introduced to identify optimal alignments between two macromolecular sequences. The algorithm guarantees discovery of the global maximum score under defined scoring matrices and gap costs.

Abstract quoted from S. B. Needleman et al., Journal of Molecular Biology (1970).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?